1. Clinical Question & Core Hypothesis

Clinical Question

TSR is a simple, reproducible H&E biomarker for stromal abundance, but it is unclear whether and how high stromal fraction causally drives poorer outcomes in MIBC versus acting as a sampling/confounding feature. The ontologist's single-path analysis frames TSR as a readout of desmoplasia, ECM remodeling, altered nutrient/oxygen diffusion, and local immune modulation?

Working Hypothesis

Primary question: In muscle-invasive bladder cancer (MIBC), does a high stromal-to-tumor ratio (TSR) measured on H&E sections independently predict poorer overall survival (OS) after controlling for standard clinico-pathologic factors and treatment? If true, the mechanism is that desmoplastic stroma (reflected by high TSR) increases interstitial pressure and ECM density, reducing chemotherapy penetration and promoting hypoxic, immunosuppressive microenvironments, thereby lowering treatment efficacy and enabling tumor progression. This yields a testable prediction: high TSR will be associated with worse OS independent of stage and grade, and this association will persist across plausible sampling strategies (e.g., including the invasive front but excluding necrotic regions). Secondary predictions include: (a) TSR remains prognostic when perioperative chemotherapy is considered, suggesting a chemo-penetration/biophysical basis; (b) the TSR–OS association is not fully explained by tumor stage or grade, supporting a mechanistic imaging biomarker role. Anchor genes remain an empty array due to data constraints, consistent with a purely imaging-driven mechanistic hypothesis. Overall, the hypothesis is structured to be testable via multivariable survival analyses and robustness checks across sampling strategies, with TSR serving as a potential marker to identify patients who may benefit from stroma-targeting or intensified systemic therapy.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Primary endpoint	OS

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	ROI_sampling_strategy (invasive_front_inclusion)	Supporting modifier from knowledge graph
Secondary	necrosis_exclusion	Supporting modifier from knowledge graph
Secondary	stain_color_normalization	Supporting modifier from knowledge graph

Secondary	therapy_type (perioperative chemotherapy)	Supporting modifier from knowledge graph
Secondary	vascular_density / perfusion proxies (if available)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ROI_sampling_strategy (invasive_front_inclusion): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- necrosis_exclusion: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.